

DAY 12: Loops

For Loop

→ a for loop is used to iterate over a sequence or iterable data types

Eg:

```
nums=[12,1,3,4,78]
for j in nums:      # ( j ) define this variable at run to store values from the
iterable datatype
print(j)
```

else in for

→ unlike if-else, the else block in for statement is executed after completed of all iterations

Eg:

```
nums='python'
for j in nums:
print(j)
else:
print('for ended')
```

break

→ The break used to stop the iteration based on the condition given

Eg:

```
nums= [1,2,3,4,5,6,7,8,9]
for num in nums:
print(num)
if num==3:
break
```

continue

→ the continue is a keyword use to skip the current iteration based on the condition

Eg:

```
nums= [1,2,3,4,5,6,7,8,9]
```

```
for num in nums:
```

```
if num==8:
```

```
    continue
```

```
    print(num)
```

```
pass
```

→ a pass is called as space holder, that is used after statements like (if , for , else) not to raise any error.

eg:

```
for i in range(1,100):
```

```
if i==20:
```

```
    print(i)
```

```
else:
```

```
    pass
```

```
assert
```

→ assert is a keyword used to check the condition , incase the condition is false, it will raise the error (AssertionError)

Eg:

```
age=15
```

```
assert age>=18, 'not eligible to vote'
```

```
print('your eligible to vote')
```

```
while loop
```

→ it is used repeatedly execute a block of code as long as a specified condition evaluates to true

if the while loop is not closed properly it will create a infinite loop

Eg:

```
num=1
```

```
while num<5:
```

```
    print(num)
```

```
    num += 1
```